

Rule Based Energy Management Strategies of Fuel Cell Hybrid Electric Vehicle

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Abstract— The outline of the paper is presenting a method to operate a Hybrid Electric Vehicle (HEV) using Fuel Cell-Battery hybrid system. The Integrated system consisting of a polymer electrolyte membrane fuel cell and a Li-ion Battery. The most common configuration for an FCHEV is the fuel cell as the main energy source, with the battery providing extra power when needed. This configuration improves efficiency and reduces the impact of high current demands. With combining PEMFC, the Integrated system output power may become governable. Two control strategies, the PID control technique and the Fuzzy logic control (FLC) technique, can be applied to the Integrated EV system. The collaboration of two control techniques, the synchronization of the PEMFC and the battery in the hybrid system and the perseverance of source variables are done. The suggested operating technique with a pliable operation mode allow always works the PEMFC with high efficiency performance band and Battery at maximum output power and thus improving the integrated system performance and magnifying system robustness. The energy management system (EMS) holds a crucial role in HEVs, serving to extend the vehicle's driving range while concurrently lowering costs. This paper introduces an energy management strategy and optimization approach for HEVs equipped with a hybrid energy storage system comprising with Li-ion battery pack and Fuel cell, implemented through a Fuzzy Logic Controller (FLC). A parametric comparison is provided with PID controller. Simulink serves as a tool for simulating the proposed system's performance, aiding in the determination of the most suitable intelligent system for achieving power efficient operation of the HEV.

Keywords— Hybrid Electric Vehicle (HEV), Fuel cell (FC), PID Controller, Fuzzy Logic Controller (FLC), Energy management system (EMS).

I. INTRODUCTION

A major downside of fossil fuel consumption is environmental pollution, which can be releases to emissions by the transportation industry. To overcome these problems, we used Energy retention-based EV, these can be categorized different types mainly, Battery electric vehicle (BEV), Hybrid electric vehicle and Plug-in hybrid electric vehicle [1]. The performance of HEV primarily relies on the effectiveness of motor drives. Due to the significant improvement in the performance of various power electronics devices such as converters, switches, and motor drives, the demand for HEV is increasing day by day [2]. A vehicle manufacturer is emphasizing HEV due to their environmental safety benefits. Hydrogen has been introduced as a fuel source for vehicles to contribute to environmental preservation. Utilizing hydrogen as the primary fuel source for electric vehicles involves the use of fuel cell technology, specifically PEMFC [3]. Hydrogen, chosen for its environmental safety and efficiency, is a crucial option for manufacturers in the realm of EV. The primary

advantage of HEV, in addition to enhanced fuel economy, lies in the reduction of carbon dioxide emissions. Fig. 1 shows the main parts of EV.

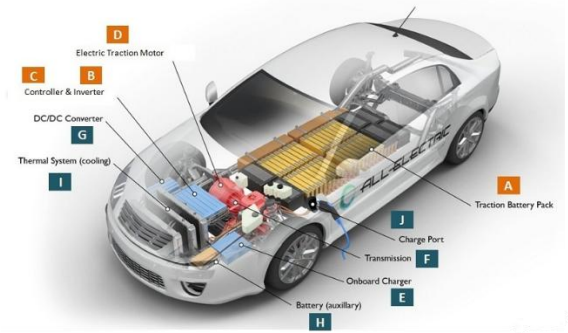


Fig. 1. Main parts of Electric Vehicle

Additionally, PEMFC boast perks such as optimal efficiency, simple implementation, and low functioning temperatures. Hence, incorporating Proton Exchange Membrane Fuel Cells in mobility solutions emerges as a promising solution to address both the ecological concerns. The key perks of Fuel Cell based Electric Vehicles encompass a substantial mileage, rapid charging times, quiet operation, and the absence of regional pollutants [4].

This paper brings forth a Simulink model encompassing the entire energy network of the powertrain in an Integrated electric vehicle, which is fuelled by both a fuel cell and a Battery. However, PEMFC, in its turn, operates most efficiently within a specific power range to bring the highest power to the load.

The lithium-ion battery serves as a power source for energy storage with rapid response capability, minimal self-discharge, extended lifespan, and substantial energy density. Consequently, the battery is commonly employed as an energetic buffer to address the shortcomings of the pure Fuel Cell Electric Vehicle [5]. A vehicle powered by both a polymer electrolyte membrane fuel cell and a battery are commonly referred to as a Fuel Cell Hybrid Electric Vehicle (FCHEV).

Modifications to EV led to the development of HEV, combining the energy sources to enhance efficiency and provide sufficient support for uninterrupted power supply [6] in vehicles. Solar and Wind generation systems are thus moving to the forefront as the primary sources of electricity generation.

Direct Torque Control (DTC) is an advanced technique recognized for its rapid torque response and robustness. It manages motor torque and flux by directly selecting the appropriate inverter switching states, bypassing the need for a

modulator or current controllers. This method is ideal for applications that demand quick torque adjustments and resilience to parameter variations. However, DTC may produce high torque ripple and necessitates complex algorithms to handle switching states effectively.

II. DIFFERENT ENERGY STORED DEVICES FOR EV

Fuel Cells (FC), Batteries, and Ultracapacitors (UC) are the most used energy storage devices for electric vehicles (EVs). These devices can function as primary energy sources in EVs or, in certain cases, serve as Indirect energy sources when operating in Integrated mode [7].

A. Fuel Cell

The FC plays a crucial role in converting chemical energy into electrical energy, serving as the primary power source in Hybrid electric vehicles. Comprising an anode, cathode, and an electrolytic membrane, the FC functions by supplying hydrogen to the anode and oxygen to the cathode. Protons move through the electrolytic membrane, while electrons pass through the load. Notably, FC stands out as an efficient energy source, contributing to a clean and emission-free power generation process [8]. Various types of Fuel Cells (FC) are employed in EV. In a hydrogen fuel cell, a catalyst at the anode facilitates the separation of hydrogen molecules into protons and electrons. These particles then follow distinct paths to the cathode. The electrons travel through an external circuit, generating an electric current or flow of electricity. This process highlights the fundamental electrochemical principles that underlie the operation of hydrogen fuel cells in EV.

PEM fuel cells boast several advantages, including high-power density and low weight. Operating at around 80°C, they facilitate quick start-ups with minimal warm-up time, making them well-suited for applications with demanding power requirements.

B. Battery

The battery stands out as an excellent and extensively utilized energy source, prevalent in virtually every electronic device [9]. It acts like a significant power source in the HEV system. The SOC represents the current level of energy stored in the battery as a percentage of its total capacity which is important tool in HEV [10]. The battery, much like Ultracapacitors (UC), plays a crucial role in providing rapid response for peak power demands during acceleration and recovering energy during deceleration.

C. UC/SC's

Supercapacitors (SC) and Ultracapacitors (UC) are recognized energy sources in HEV systems, distinguished by variations in electrode length and storage capacity. In the case of supercapacitors, they function by storing electrical energy between two electrostatic double layers formed through the creation of thin charge layers at the interface of the electrolyte and electrodes [5]. Ultracapacitors differ from conventional batteries by not storing electrons directly; rather, they store positively and negatively charged ions, utilizing a liquid electrolyte to enable energy flow. Unlike batteries, ultracapacitors do not involve chemical reactions in their operation.

III. SYSTEM DESCRIPTION

The Integrated system comprises a Battery-Fuel Cell source connected to loads, including an electric vehicle. The

key energy sources in this hybrid setup are the Battery and the Fuel cell. To facilitate the integration of these sources, dc-dc converters are employed. This arrangement allows for efficient conversion and management of electrical power between the Battery-FC hybrid source and the loads, including the electric vehicle, contributing to a comprehensive and well-integrated hybrid energy system [11].

From Fig. 2 FC and output from the battery system commonly connected to Buck/Boost converter. Output of the Buck/Boost converter is connected to the motor which is connected to the wheels of an electric vehicle. Where fuel cell discharges at the time of accelerating condition and at the time of regenerative braking battery stores energy. At the condition of starting of vehicle it requires full acceleration then both battery and fuel cell will work, after reaching to accelerated point for smooth operation fuel cell will works and battery will charge.

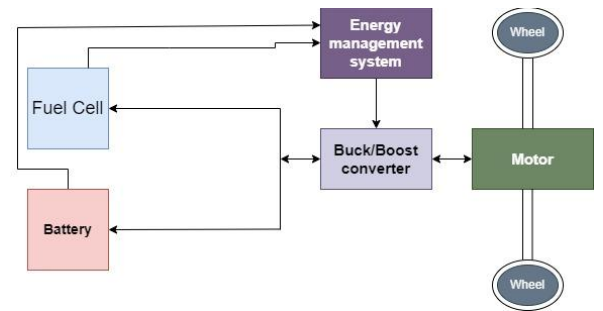


Fig. 2. Electric vehicle connected Hybrid system

The energy management system is crucial for maintaining the SOC of the battery in an electric vehicle. By regulating the torque percentage within a specific range based on the vehicle's speed, the SOC can be optimized. This can be achieved using a FLC, which is designed to manage the SOC effectively. While a PID controller is also commonly used in electric vehicles to maintain SOC and enhance driving range, the extended version of this system, known as the Fuzzy Logic Controller, will be compared in the results section.

IV. OPERATING STRATEGY OF THE HYBRID SYSTEM

The polymer electrolyte membrane fuel cell system, serving as the original power source in the Fuel Cell based Hybrid Electric Vehicle, comprises the fuel cell stack, supporting system, and dc/dc converter. The generated power of the polymer electrolyte membrane fuel cell system can be calculated by

$$P_{FC,sys} = (P_{stack} - P_{aux}) \eta_{DC/DC} \quad (1)$$

$$\text{Where,} \quad P_{stack} = f(I_{FC}) \quad (2)$$

Where P_{aux} denotes auxiliary system power consumption (the unit is kW), P_{stack} is the fuel cell stack output power (the unit is kW), $\eta_{DC/DC}$ is the dc/dc converter efficiency, and I_{FC} represents the fuel cell current (the unit is A).

The PEMFC system efficiency is,

$$\eta_{FC,sys} = \frac{2FV_{FC}I_{FC}}{N_{FC}I_{FC}\Delta H_{H_2}} \left(1 - \frac{P_{aux}}{P_{stack}}\right) \eta_{DC/DC} \quad (3)$$

F is the Faraday's constant, VFC indicates the fuel cell induced voltage, NFC represents the quantity of cells, and ΔH_{H_2} denotes the hydrogen higher heating value.

The hydrogen consumption mass of fuel cell system is,

$$M_{H_2} = \int_0^T C_{FC} dt \quad (4)$$

Where,
$$C_{FC} = \frac{P_{FC,sys}(t)}{\eta_{FC,sys} \Delta H_{H_2}} \quad (5)$$

Here, C_{FC} represents the hydrogen usage rate of the studied stack and M_{H_2} is the hydrogen consumption mass of the polymer electrolyte membrane fuel cell.

Moreover, the coefficient K_{FC} is helped to constrain the stack to function within the designated high-efficiency operational range. It shown in Fig. 3.

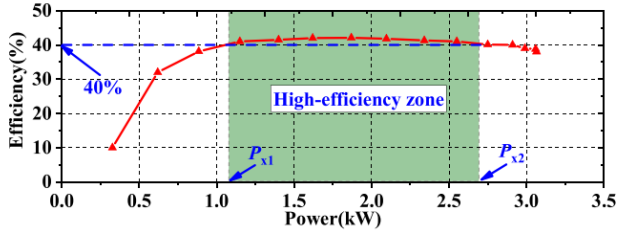


Fig. 3. Polymer electrolyte membrane fuel cell high-efficiency operating zone

In Fig. 3, The limits of the high-efficiency operational range for the utilized system polymer electrolyte membrane fuel cell, denoted as P_{x1} and P_{x2} are variable parameters. Real-time updates for these two parameters are essential and can be achieved through the utilization of the Extended Kalman Filter (EKF) algorithm. The EKF algorithm facilitates dynamic and real-time estimation, allowing for the continuous adjustment and optimization of P_{x1} and P_{x2} based on changing operating conditions and system dynamics. This adaptability ensures that the Polymer electrolyte membrane fuel cell operates within its high-efficiency range, contributing to overall system performance and efficiency.

In the hybrid system, the battery functions as the energy storage unit, serving the purpose of maintaining the bus voltage and capturing regenerative braking energy. The Battery Management System (BMS) monitors and controls crucial operating parameters of the battery. Among these parameters, the State of Charge and essential factors for hydrogen consumption equivalence for the Energy Management System [12]. The battery's SOC can be calculated using the subsequent equation,

$$SOC = SOC_{init} - \frac{1}{Q} \int_0^T I_{bat} dt \quad (6)$$

Where SOC_{init} is the initial value of SOC, Q_{bat} is the capacity of battery, and I_{bat} is the battery outcome current. In addition, accordance with the experimental test, the correlation between the VOC and the SOC is demonstrated in Fig. 4(a).

Furthermore, the internal resistance of the battery is intricately connected to the State of Charge, as demonstrated in Fig. 4(b).

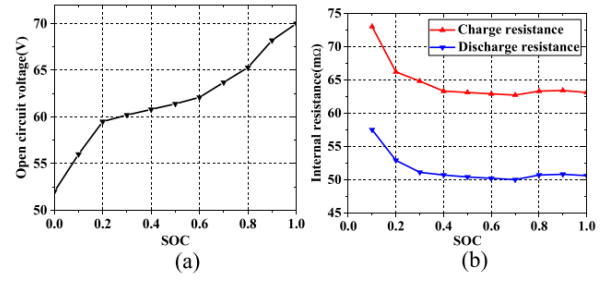


Fig. 4. The battery's characteristics. (a) V_{OC} . (b) Internal resistance

V. FUZZY LOGIC CONTROLLER FOR HYBRID SYSTEM

The primary application of the FLC is in managing various converters, and its functionality has now expanded to regulating the SOC in HEV. The efficiency of the controller is influenced by several system variables, making it crucial for the controller to manage all parameters in synchronization with changes in the battery status. In such cases, the Fuzzy Logic Controller proves to be the optimal solution, showcasing its ability to effectively regulate the SOC of the system.

The primary function of the FLC involves selecting the appropriate input and output variables, as well as defining the membership functions [13]. These functions can vary and are used in max-min composition, alongside AND/OR rules, to generate the controller's output. This process ensures that the output of the controller accurately reflects the system's behavior by considering the relationships between the input variables.

Fig. 5 shows the working diagram of FLC. The SOC and power serve as the input variables for the FLC. The process begins with fuzzification, where the fuzzifier converts the input values into fuzzy sets. Next, the fuzzy inference system applies predefined rules to process these fuzzified inputs and generate an intermediate output. Finally, this output is sent to the defuzzification block, which produces the current as the final, crisp output of the system [14].

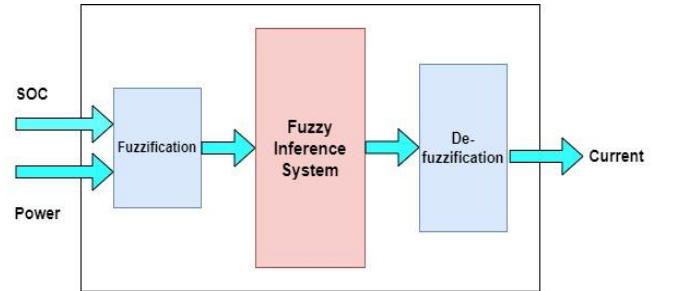


Fig. 5. Working of Fuzzy logic controller for HEV

The key advantage of such systems lies in their adherence to precise rules, ensuring that operations are carried out within the defined rule limits. By utilizing the current SOC of the battery along with the available generator power, the system can optimize the SOC profile. This, in turn, enables the vehicle to operate in hybrid mode for extended periods, supporting longer-distance travel [15].

The SIMULINK model used to simulate Fuzzy logic controller is represented in below figure 6.

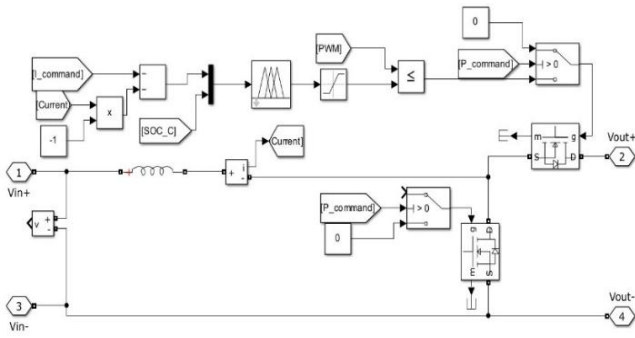


Fig. 6. Simulink model of Fuzzy controller: Buck/Boost converter

VI. RESULTS AND DISCUSSIONS

At the time of starting and full running of EV it requires higher power so at this condition Electric vehicle requires higher power demand. So, at the time of starting both battery and fuel cell will work and generate the power to start and run the vehicle. Simulation results are shown for both type of controllers in Fig. 7 at the condition of high power demand. Fig. 7(a) and Fig. 7(b) shows the results Battery SOC using PID controller and FLC controller. From Figure 7(a), the SOC curve initially at 100% and smooth till 10 sec after that it gradually decreases to 92% in 100 seconds with PID controller. SOC reaches to 92% at the conclusion of the drive cycle. From Figure 7(b), the SOC curve of battery with Fuzzy controller is initially at 100% and gradually decreases to 99.95% at 20sec. From 20sec to 50sec SOC maintained smoothly to 99.94%. Finally, Battery slowly discharges to 99.91% at 100sec.

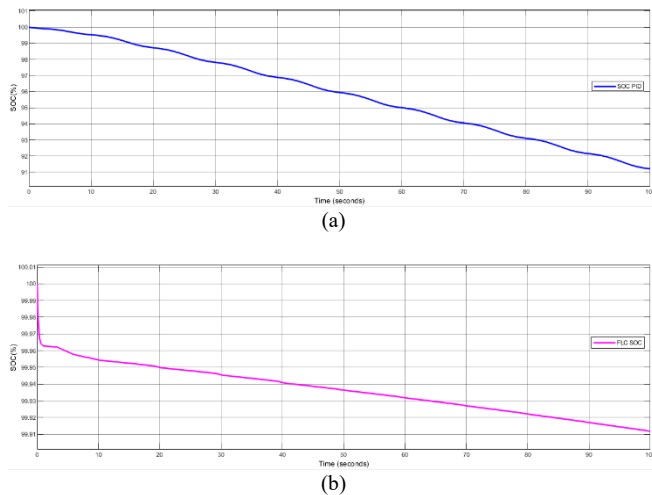


Fig. 7. (a) Battery SOC using PID controller, (b) Battery SOC using FLC controller

Fig. 8 indicates the Electric Vehicle's Speed and Motor Torque results. Initially, the motor generates high torque with low speed as the speed stabilizes, the torque gradually decreases.

At the time of braking condition, the motor will work as a generator and charges the battery using regenerative braking method.

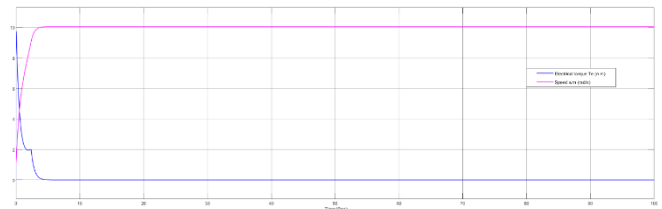


Fig. 8. indicates the Electric Vehicle's Speed and Motor Torque

Table 1 below presents a comparison of various parameters between the PID controller and the Fuzzy Controller.

Table 1. Comparison of various parameters using different Energy management systems

Parameter	Battery Controller	
	PID	FLC
Time(sec)	100	100
Initial SOC%	100	100
Final SOC%	91	99.94
SOC drop	Sharp	Small

The findings indicate that the Fuzzy logic controller is a more advanced and precise system, grounded on the input/output data pairs of the system.

VII. CONCLUSION

A hybrid energy management system combining PID and FLC methodologies is implemented to efficiently regulate power distribution in a HEV with battery-based energy storage. The PID controller manages the battery SOC, optimizing it for longer journeys, while the FLC system, utilizing input/output data pairs, fine-tunes SOC control. Advanced optimization techniques are applied to both the PID and FLC systems, enabling a comparative analysis of their respective performances.

The simulation results show that the PID controller maintained the State of Charge SOC at 91% by the end of the drive cycle, particularly during accelerations. In contrast, when using the FLC with a Triangular membership function, the battery SOC remained above 99.94% at the end of the drive cycle. This highlights the superior performance of the FLC in maintaining SOC, as it achieved approximately 99.94% even after a 100-second drive, demonstrating its effectiveness in optimizing battery energy management.

HEV represent the future of transportation, focusing on the integration of intelligence and power efficiency in their design. This study contributes to the advancement of controllers that satisfy both essential criteria, helping pave the way for a smoother transition to sustainable transportation solutions.

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